Draft version July 18, 2026
Typeset using LaTeX default style in AASTeX63

Measuring the footprint of large language models in the astronomy literature

Serat Mahmud Saad[1]

[1] Department of Astronomy, The Ohio State University, Columbus, OH 43210, USA

ABSTRACT

Large language models became easy to use in late 2022, and their use in scientific writing has grown quickly in most fields, but the size of this change in astronomy has not been reported. We measure it from 200,547 astro-ph abstracts posted to arXiv between 2015 and 2026. The fraction of abstracts that contain a "marker" word, rare before ChatGPT and common after, rose from 1.5% before 2022 to 5.8% in 2025, while a matched set of neutral words stayed flat. A conservative lower bound puts the model-touched fraction of 2025 abstracts near 4%, robust to baseline and word-list choice. Chosen without an imported list, the data recover the same style words next to the genuinely new instrument names, and the marker words occur together far more often than chance (27 of 91 pairs above independence at $p < 0.01$). Against these numbers, explicit disclosure of model use stays below 1% and direct checks of the LaTeX source find near-zero pasted text. All 22,547 cited arXiv identifiers in 2,532 recent papers resolve, so fabricated citations, which are rising in biomedicine, sit below a 0.24% per-paper limit here. The word signal weakens in 2026 even as disclosure rises. All data and code are public.

Keywords: Astronomy data analysis — Scholarly communication — Natural language processing

1. INTRODUCTION

The public release of ChatGPT in November 2022 made large language models easy to use, and their use in scientific writing followed quickly (Stokel-Walker & Van Noorden 2023; Van Noorden & Perkel 2023). Model-written scientific text is hard to tell from human writing even for expert readers (Else 2023), and journals responded early with authorship bans and disclosure rules (Nature 2023). Detecting model use in any single document is unreliable: commercial detectors disagree with each other, are simple to evade with light editing, and misclassify text by non-native English speakers at high rates (Liang et al. 2023). Recent work therefore measures the effect at the level of a whole corpus, by tracking words whose frequency changed abruptly after 2022. The approach works because the models themselves have measurable lexical preferences. Words such as `delve` and `intricate` appear in model output at many times their human rate, an overproduction that Juzek & Ward (2024) trace in part to the human-feedback tuning stage rather than to the training corpus alone. When even a modest fraction of authors pass their text through the same few models, the shared preferences of those models become visible in the aggregate vocabulary of a field. Liang et al. (2025) fit a statistical model to arXiv, bioRxiv, and the Nature journals and report that up to about 17.5% of computer science abstracts by early 2024 carried model-edited text. Kobak et al. (2025) count "excess" vocabulary in more than 15 million PubMed abstracts and find that at least 13.5% of 2024 abstracts were processed by a model. Gray (2025) estimate that a model was involved in more than 10% of all 2024 papers and describe a cat-and-mouse effect in which words lose their signal once they become known. Corresponding signals appear in article full text across fields (Thelwall & Kousha 2026) and in conference peer reviews, where 6.5–16.9% of reports were substantially model-modified within a year of ChatGPT's release (Liang et al. 2024). Physics and mathematics sit lower in these studies, near 6–10%.

Astronomy has been studied less, although the tools have entered its daily practice, from conversational interfaces to the literature (Ciucă & Ting 2023) and semantic literature search (Iyer et al. 2024) to domain-tuned models and benchmarks (Perkowski et al. 2024; Ting et al. 2024), alongside code assistance and copy editing. An early keyword estimate placed model involvement at about 1% of all 2023 papers (Gray 2024). Astarita et al. (2024) count model-favored words across about one million records in the NASA Astrophysics Data System (Kurtz et al. 2000) and show a clear rise after 2023, larger in non-refereed work; they do not convert the signal into a fraction. Geng & Trotta (2025) report the cat-and-mouse effect across all of arXiv. A survey of astronomers (Fouesneau et al. 2024) finds that most use these tools every week but few state so in print. No paper has reported, for astronomy specifically, how large the footprint is, how it compares with other fields, or how the different ways of detecting it agree.

The size of the effect matters for practical reasons. If a large share of abstracts are edited by a model, then word-based screening of papers or proposals will flag honest authors, and in particular non-native English speakers, at high rates. If model use is common but rarely stated, a reader cannot tell how a given text was produced. And if models draft text rather than only polish it, the record may drift toward a narrower and more uniform style. It is also useful to separate three uses that are easy to run together: a model as a copy editor, a model that drafts content, and a paper produced whole. The methods here mostly measure the first, and a measurement of its size is a first step toward any of these questions.

In this paper we harvest every astro-ph abstract from 2015 to 2026 and measure the marker-word signal, its rise, and its recent weakening. We convert the signal into a lower bound on the fraction of abstracts a model touched. We let the data select the words and two-word phrases without an imported list, and we measure how often the marker words occur together. We then compare the word signal against three other measurements on the same corpus: explicit disclosure in acknowledgments, pasted-model text in the LaTeX source, and fabricated citations. We close with the dependence on subfield and author country, a measurement of promotional versus hedging language, and the limits of the method. Section 2 describes the data and methods, Section 3 presents the results, and Section 4 the discussion; the appendix after the references lists the full word basket. Throughout, a marker word is treated as a sign of model-edited language, not of poor or dishonest science.

2. DATA AND METHODS

2.1. *Corpus*

We queried the arXiv API for every paper with a primary or cross-listed astro-ph category from January 2015 through July 2026. The harvest was run in month-sized date windows, which keeps each query below the interface's pagination limits and makes the download repeatable; papers appearing under several versions were reduced to one record keyed on the base identifier. We kept the version-1 submission date, the primary category, and the abstract, which gives 200,547 abstracts. We work with abstracts rather than full text for three reasons: they are available uniformly for every paper over the whole period, they are the part of a paper most likely to receive careful editing, and they are the unit used by the studies we compare against (Liang et al. 2025; Kobak et al. 2025). The API returns the abstract of the latest posted version, so a paper revised after 2022 can carry recent language back to its original submission date. This smears the signal backward in time and biases the measured rise down; the flat pre-2022 marker rate in Section 3.1 shows the effect is small. The 2026 bin covers January through July. We tokenize each abstract into lowercase alphabetic words. A word's "document frequency" in a year is the fraction of that year's abstracts that contain the word at least once, which is the quantity used in earlier work.

For disclosure (Section 3.5) and for the by-country measurement (Section 3.7) we use the NASA Astrophysics Data System, whose `ack` field searches the acknowledgments section and whose `full` field searches full text where available. We count `database:astronomy` records per year as the denominator.

2.2. *Marker, control, and tone vocabularies*

We assembled a basket of 38 marker words from earlier studies (Kobak et al. 2025; Liang et al. 2025; Geng & Trotta 2025): `delve`, `underscore`, `intricate`, `pivotal`, `showcasing`, `meticulous`, and related forms; the full list is given in Appendix A. Each entry satisfies two conditions: it is reported as model-favored in at least one prior corpus study, and its document frequency in astro-ph before 2022 is low, mostly below 0.2% of abstracts. The second condition matters for the estimator of Section 2.5, because a word that is already common in human writing contributes noise rather than signal. The basket is chosen for low baseline and high specificity rather than for size; broader baskets give larger signals but add words with mixed meanings. As a control we use ten neutral astronomy words (`observed`, `measured`, `galaxy`, and others listed in Appendix A), selected to be common, stable in frequency over 2015–2022, and spread across observation, analysis, and subject terms, so that any corpus-wide change in abstract style would move them as well. To measure tone (Section 3.8) we also track 24 promotional words (`unprecedented`, `remarkable`, `striking`, and similar) and 21 hedging words (`may`, `might`, `suggest`, and similar) that mark stated caution; the full sets are in the data release.

2.3. *Data-driven vocabulary selection*

Because the marker basket is imported from other fields, we also select vocabulary from the astronomy corpus alone. For every word, and separately for every two-word phrase, we compute the document frequency in a pre-ChatGPT baseline window (2018–2021) and in a recent window (2024–2026), keep items with at least 30 occurrences (25 for phrases) in one of the two windows, and rank by the ratio of the two frequencies. We rank by ratio rather than by absolute frequency change because the model-favored words start from very low baselines; an absolute-change ranking is dominated by common words whose fractional change is small. For the classified tail shown in the figures we require in addition a baseline frequency above 0.05%, so that a ratio is measured rather than divided by a near-zero count, and we keep a small set of well-known rare markers (`delve`, `delves`, `tapestry`) that fall below that floor. No external word list enters this selection.

2.4. *Co-occurrence statistics*

If the marker words share a common cause, they should occur together within abstracts more often than chance. For a pair of words a and b we measure the lift

$$\mathrm{lift}(a, b) = \frac{P(a \cap b)}{P(a)\,P(b)}, \tag{1}$$

where $P(a)$ is the fraction of 2024–2026 abstracts that contain word a and $P(a \cap b)$ the fraction that contain both; a lift of one means the two words are independent. We evaluate the lift for the 91 pairs formed by the 14 most frequent basket markers. Because several markers are rare, a large lift can rest on a handful of abstracts, and a χ^2 test is not valid at expected counts far below one. We therefore test each pair against a Poisson null: with $N = 53{,}192$ recent abstracts the expected joint count under independence is $\mu = N\,P(a)\,P(b)$, and we compute the probability of the observed count or more. With 91 tests at $p < 0.01$, about one false positive is expected by chance, which sets the scale against which the observed number of significant pairs should be read.

2.5. *Lower-bound estimator*

To convert the signal into a fraction we count the share of abstracts that contain at least one basket word and subtract the pre-2022 rate. Let f_t be the fraction of abstracts in period t that contain at least one basket word and let f_0 be the same fraction in the 2018–2021 baseline. Our estimate is

$$\alpha_{\mathrm{lb}} = f_t - f_0. \tag{2}$$

This is a lower bound for two reasons: a model-edited abstract need not keep any word from a short list, so f_t misses some model-touched abstracts; and by subtracting f_0 we discard any human writer who uses a basket word for ordinary reasons. Both effects push the estimate down. The estimator is the document-level counterpart of the excess-vocabulary measure of Kobak et al. (2025), and it is more conservative than the mixture fit of Liang et al. (2025), which models the full word-frequency distribution and therefore counts model-edited abstracts that keep none of the listed words; this difference should be kept in mind when the resulting numbers are compared. Equation 2 is the $q = 1$ limit of a two-population model: if a fraction α of abstracts are model-edited and carry a basket word with probability q, while the rest carry one at the baseline rate f_0, then $f_t = (1 - \alpha)f_0 + \alpha q$ and $\alpha = (f_t - f_0)/(q - f_0)$. Since $q \leq 1$, $\alpha \geq f_t - f_0$, and any realistic $q < 1$ raises α above the bound.

2.6. *Disclosure searches*

We search astronomy records in the NASA Astrophysics Data System for terms that identify model use without ambiguity: ChatGPT, GPT-4, "large language model", and GitHub Copilot. We exclude terms that are ambiguous in astronomy, such as "Gemini" (the observatory) and "Claude" (a personal name); the excluded terms match hundreds of astronomy papers per year through the Gemini telescopes alone, so this cut is not optional. Acknowledgment matches are read as stated use; full-text matches include discussion of models as a research topic and give an upper envelope of mentions. To check what the acknowledgment field captures, we read 25 randomly drawn matched acknowledgments from 2024: all 25 are ordinary astronomy papers stating that a model was used for language or code, and none is a paper about language models.

2.7. *Source-text audit*

arXiv distributes the LaTeX source of most papers, and model boilerplate is sometimes pasted into the source in comments that never reach the compiled document, such as "as an AI language model"; a growing catalog of such phrases surviving into published articles exists for other fields (Glynn 2024). We downloaded the source of 3,950 papers, sampled with a fixed stride within each year from 2016 to 2026; about 87% have recoverable text source. We search two pattern classes: hard boilerplate that has no legitimate reason to appear in a manuscript (refusal phrases, "as an AI language model", knowledge-cutoff statements), and mentions of specific tools inside comments. Class and style files are excluded, since package boilerplate produces false matches, and non-text payloads are filtered before searching. The patterns were tuned on the pre-2023 papers, which contain no model text by construction and therefore measure the false-positive rate of the search directly.

2.8. *Citation audit*

Models sometimes produce citations to works that do not exist. Together with industrial-scale paper mills (Richardson et al. 2025), such references are a growing integrity concern, and their rate is rising in biomedicine, from about one paper in 2828 in 2023 to one in 458 in 2025 and one in 277 in early 2026 (Topaz et al. 2026). To test for this failure mode in astronomy we extracted every cited arXiv identifier and digital object identifier (DOI) from the LaTeX source of 2,532 papers from 2022–2026, drawn with a fixed stride within each year and weighted toward 2024–2026. Every new-style arXiv identifier is checked against the arXiv API; a random sample of 2,500 of the 56,706 unique DOIs is checked against Crossref, and every unresolved case is inspected by hand against the global handle resolver and the citing reference. We quote two statistics: a per-paper rate, defined over papers with at least one checkable identifier and directly comparable to the biomedical audit of Topaz et al. (2026), and a per-identifier rate. Upper limits for zero detections follow the usual rule of three, $3/N$ at 95% confidence.

3. RESULTS

3.1. *The rise of the marker vocabulary*

Figure 1 shows the yearly document frequency of the marker basket and of the control set. The marker basket sat between 1.0% and 2.0% over 2015–2022, with a slow drift of 0.12 percentage points per year that we return to in Section 3.4, and rose to $5.80 \pm 0.16\%$ by 2025. Against the pooled 2018–2021 baseline of 1.50%, the 2025 rate differs by $z = 35$ in a two-proportion test. The mean frequency of the control set changed by +0.10 percentage points between the same two periods, forty times less than the basket change, although individual control words drift by up to 2 percentage points for topical reasons (`obtained` fell 2.1 points; `redshift` rose 1.7). The rise is specific to the model-favored vocabulary and is not a general change in how abstracts are written.

The aggregate rate conceals substantial evolution among the contributing words. Figure 2 tracks individual words by quarter. The quarterly basket rate climbs from about 2% at the start of 2022 to a maximum of 6.1% in the first half of 2025. The frequency of `delve` rose from below 0.05% to a peak of 0.29% of abstracts in the first quarter of 2024 and then returned to its pre-2022 level by 2026, after the word received wide public attention as a sign of model text. Less publicized words behaved differently over the same interval: `underscore` and `leveraging` continued to rise through 2025, the latter reaching 1.5% of abstracts. The total marker rate did not fall with `delve`; the decline was offset by other words. Public attention to a specific marker removes that marker without reducing the aggregate signal, which matches the cross-field result of Geng & Trotta (2025).

3.2. *Data-driven vocabulary and phrases*

The ranking of Section 2.3 provides a check that does not depend on the imported basket. The top of the ranking splits into two families (Figure 3). One family consists of instruments and surveys that began operating during the recent window (JWST's NIRCam, MIRI, and NIRSpec; the DESI survey; XRISM), as expected for genuine changes in research topics. The other family is the same style vocabulary reported in computer science and biomedicine, recovered here independently from the astronomy corpus: `leveraging`, `pivotal`, `intricate`, `highlighting`. Table 1 lists a representative set with document frequencies before and after ChatGPT.

The same separation holds for two-word phrases. Figure 4 tracks a set of model-favored bigrams over time; `underscores the` and `highlighting the` rose by an order of magnitude, while a neutral control phrase (`wide range`) stayed flat. Ranking all bigrams by their frequency ratio again splits topic phrases (`jwst observations`, `the desi`) from style phrases (`offering a`, `results highlight`, `findings indicate`), as shown in panel (b) of Figure 3. Phrases carry a stronger signal than single words, and they are harder to remove by editing one word at a time.

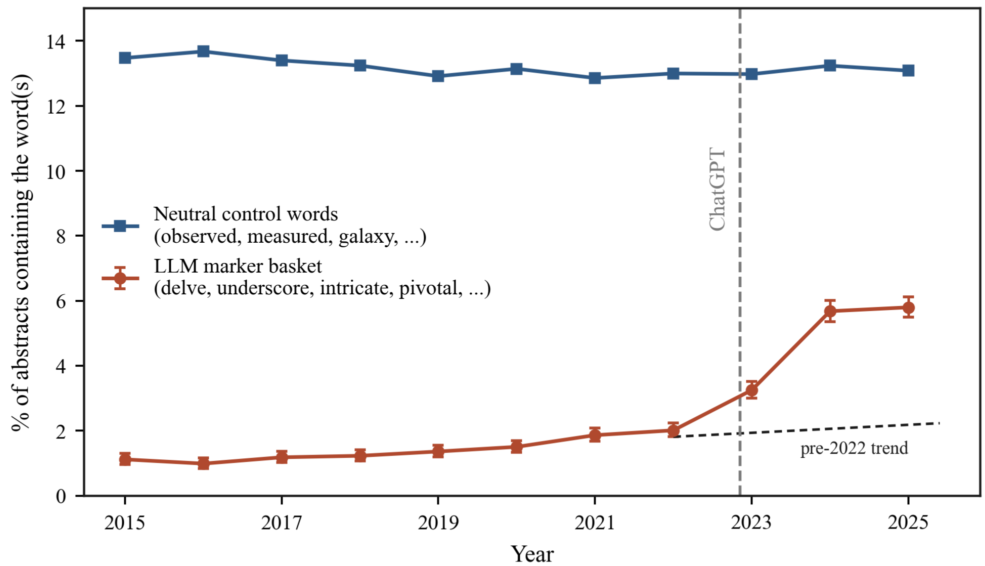

Figure 1. Fraction of astro-ph abstracts containing a marker word (red) and a matched set of neutral control words (blue), by year. Error bars on the marker curve are Wilson 95% intervals; most are comparable to the symbol size ($n \simeq$ 15,000–22,000 abstracts per year). The black dashed segment extrapolates the 2015–2021 linear trend; the excess over it after 2022 is the signal. Both curves are per-abstract fractions, so we stop at the last full year, 2025.

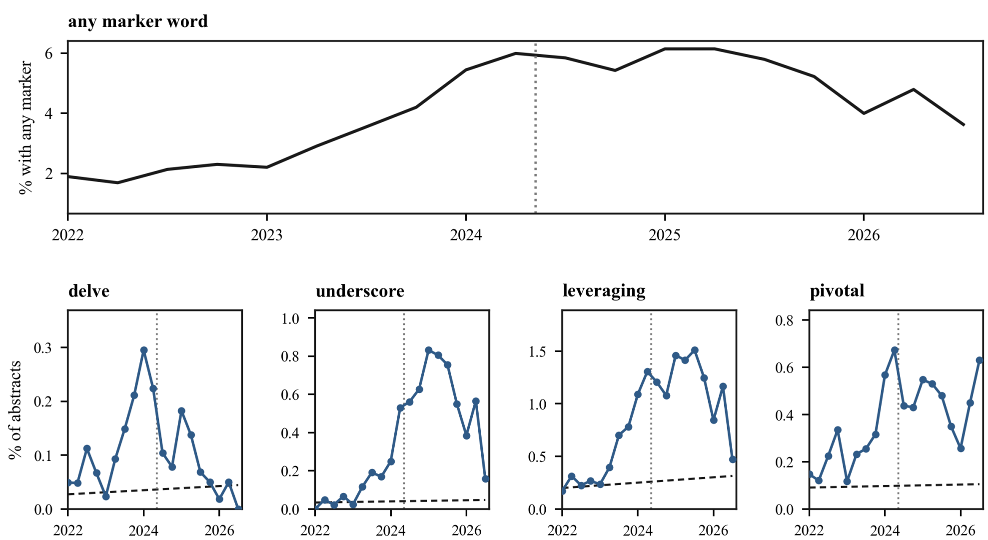

Figure 2. Top: the fraction of abstracts with any marker word, by quarter. Bottom: four individual words, each on its own scale; the dashed black segment in each panel extrapolates that word's 2015–2021 linear trend. The vertical dotted line marks mid-2024, when `delve` received wide public attention as a sign of model-generated text; its frequency declines from that point while the other markers continue to rise, and the aggregate rate does not fall.

3.3. *Co-occurrence of marker words*

Of the 91 marker pairs, 27 exceed the independence expectation at $p < 0.01$, where 0.9 false positives would be expected; Figure 5 shows the significant cells. Among pairs with at least five joint occurrences the lift runs from 3 to 36 (`delves`+`intricate`: lift 36 from 5 joint abstracts against an expectation of 0.14, $p = 4 \times 10^{-7}$; `intricate`+`pivotal`: lift 10 from 10 against 0.97, $p = 8 \times 10^{-8}$). The largest lifts among the rarest words rest on one to three abstracts each and should be read as order-of-magnitude values, although even a single joint occurrence is significant when the expectation is 2×10^{-3}. The mean pairwise lift is 14, and 11 when restricted to pairs with at least five joint occurrences.

Table 1. Representative marker words: document frequency in the pre-ChatGPT baseline (2018–2021) and the recent window (2024–2026), and the ratio of the two.

word	2018–21 (%)	2024–26 (%)	ratio
`underscore`	0.03	0.58	23
`showcasing`	0.01	0.12	10
`leveraging`	0.13	1.22	9
`offering`	0.26	2.08	8
`delve`	0.02	0.11	7
`pivotal`	0.08	0.47	6
`highlighting`	0.34	1.68	5
`intricate`	0.07	0.39	5

NOTE—Document frequency is the fraction of abstracts that contain the word at least once. The full table of every word that clears a count floor is in the public data release.

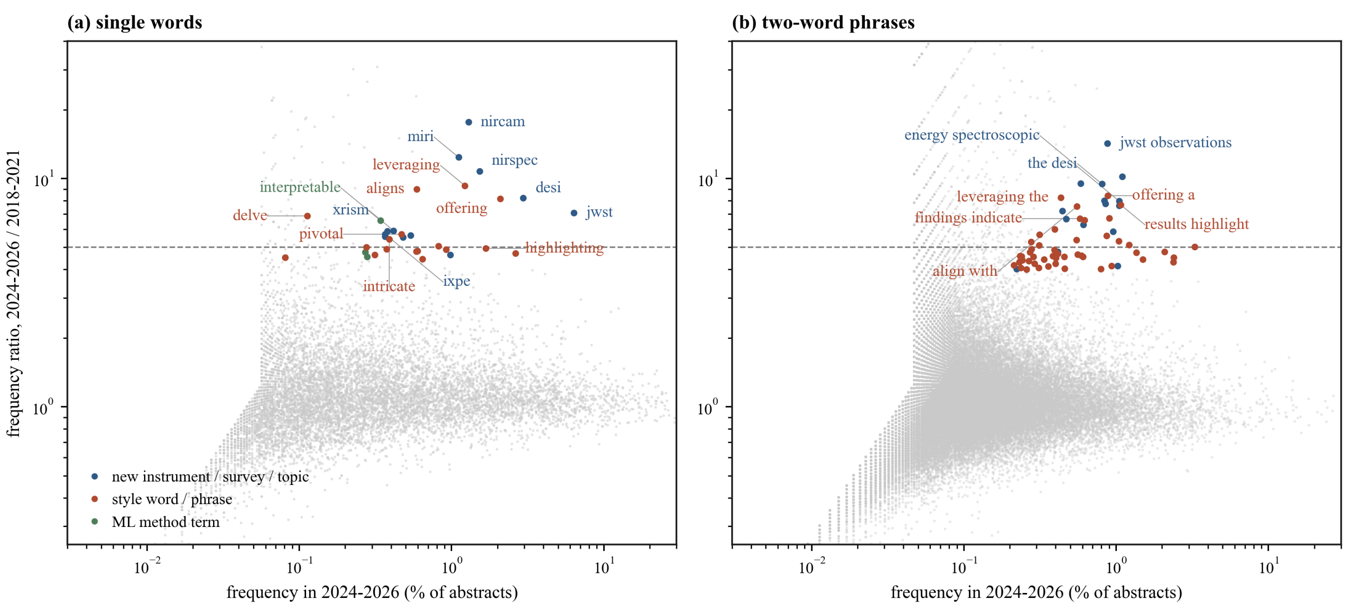

Figure 3. Frequency ratio between 2024–2026 and 2018–2021 against recent frequency for (a) every word and (b) every two-word phrase in the corpus (grey points). The tail above the dashed line at ratio 5 splits into new instruments, surveys, and topics (blue), style words and phrases (red), and technical machine-learning terms (green); selected items are labeled. Axes are logarithmic.

Independent drift of unrelated words would not produce this pattern, whereas a model that edits or drafts a whole abstract, and adds several of its preferred words at once, would. Part of the clustering could be topical, since marker use is not uniform across subfields (Section 3.7), but the strongest pairs join words with no shared subject matter. The clustering therefore supports reading the marker signal as one effect rather than a set of unrelated trends.

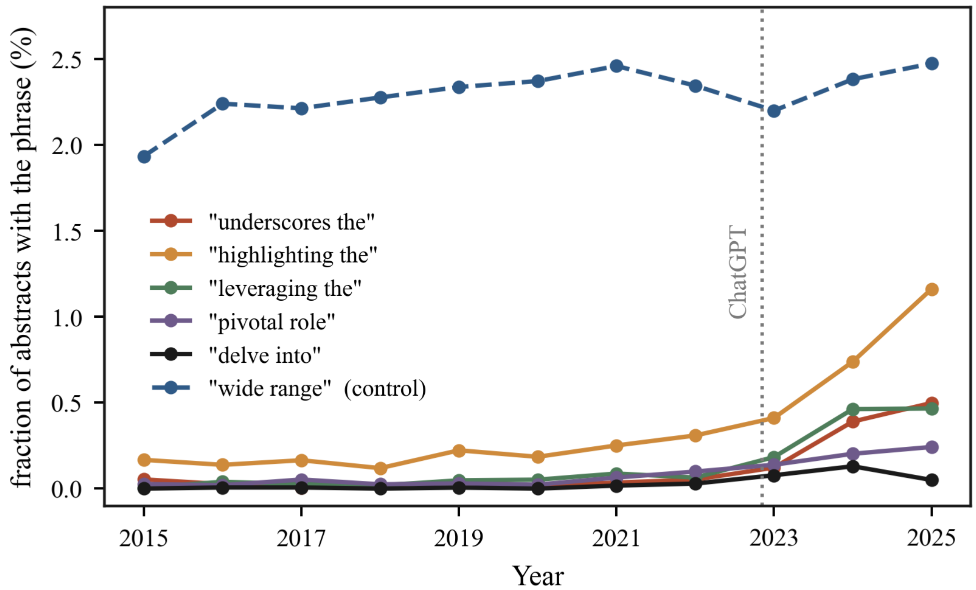

Figure 4. Fraction of abstracts containing selected two-word phrases, by year. Model-favored phrases rise after 2022; a control phrase (`wide range`) stays flat.

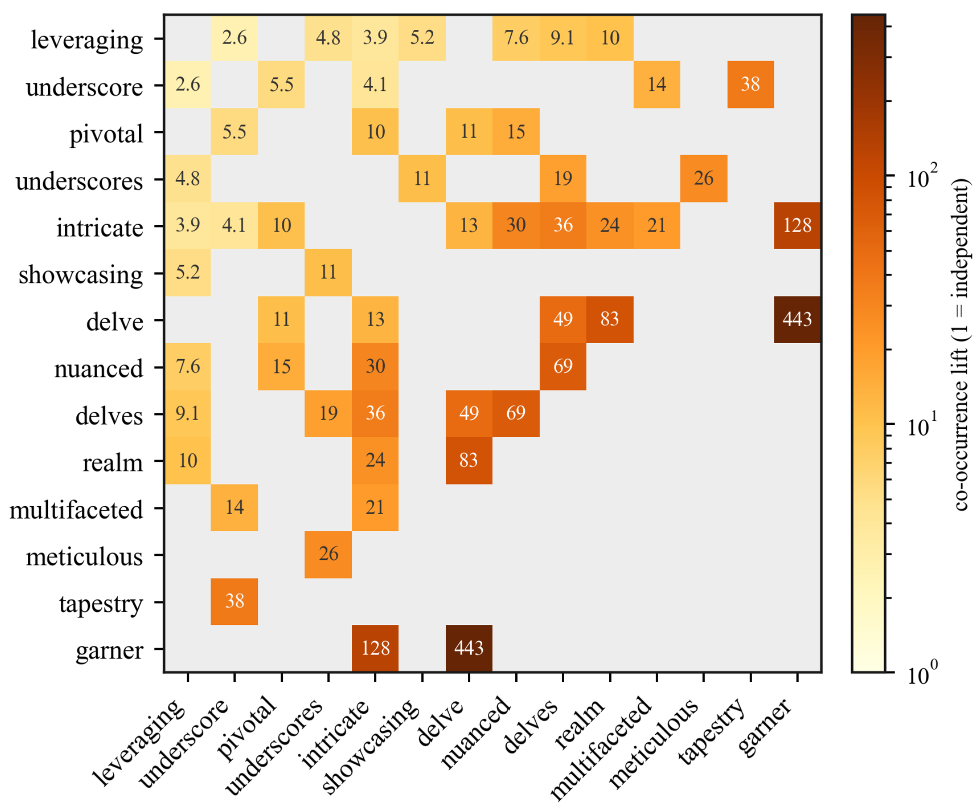

Figure 5. Co-occurrence lift for pairs of marker words in 2024–2026 abstracts. Colored cells are the 27 of 91 pairs whose joint count exceeds the independence expectation at $p < 0.01$ (Poisson test); grey cells are consistent with independence. Lifts above $\sim$40 rest on one to three joint abstracts and are order-of-magnitude values.

Table 2. Robustness of the lower bound $\alpha_{\rm lb}$, pooled 2024–2026 window.

variant	$\alpha_{\rm lb}$ (%)
default (baseline 2018–2021)	3.9
baseline 2017–2020	4.1
baseline 2016–2019	4.2
baseline 2015–2018	4.3
worst leave-one-word-out (`leveraging` removed)	3.0
control words, same estimator	0.1

NOTE—For 2025 alone the default gives $4.30 \pm 0.16\%$, and a linear extrapolation of the pre-2022 drift in place of the flat baseline gives 3.6%.

3.4. *The lower bound and its robustness*

For 2025 the estimator of Section 2.5 gives $f_t = 5.8\%$ and $f_0 = 1.5\%$, so $\alpha_{\rm lb} = 4.30 \pm 0.16\%$, where the error is the binomial error on the two proportions; the true value is higher.

Table 2 collects the variations we tried, computed for the pooled 2024–2026 window. Moving the four-year baseline window earlier raises the pooled bound by at most 0.36 percentage points, because the basket rate drifted up slowly even before ChatGPT (0.12 points per year over 2015–2021). Treating that drift as a secular trend and extrapolating it linearly raises the effective 2025 baseline to 2.2% and lowers the 2025 bound from 4.3% to 3.6%, which we consider the most conservative reading. Removing any single word from the basket changes the pooled bound by at most 1.0 point; the largest single carrier is `leveraging`, without which it is 3.0%. The control-word set run through the same estimator gives +0.10 points. Under every variant the bound stays between 3.0% and 4.3%.

Figure 6 places this number against published estimates for other fields: 17.5% for computer science abstracts and 6.3% for mathematics and the Nature portfolio (Liang et al. 2025), and at least 13.5% for biomedical abstracts (Kobak et al. 2025). Our bound of 4.3% sits below computer science and biomedicine and closest to the mathematics value. Because our number is a strict lower bound while the other studies use fuller estimators that also count abstracts that keep none of the listed words, the comparison compresses the true astronomy value from below, and we conclude that astronomy is consistent with the range measured for the rest of science, at a few percent to low double digits.

3.5. *Stated use*

The count of astronomy papers matching the disclosure terms of Section 2.6 is exactly zero before 2023, which confirms that the search itself is clean, and rises every year after (Figure 7). As a share of all astronomy papers it stays below 1%: about 0.20% of 2025 papers state model use in the acknowledgments and about 0.67% mention a model anywhere in the text; the inspected acknowledgment matches (Section 2.6) confirm that these are genuine statements of use. Full-text coverage in the database is incomplete, so these fractions are themselves lower limits, but the missing coverage would have to be large and heavily skewed to close the factor-of-twenty gap to the word signal. Set against the 4% floor from the word counting, model use is common and stated use is rare.

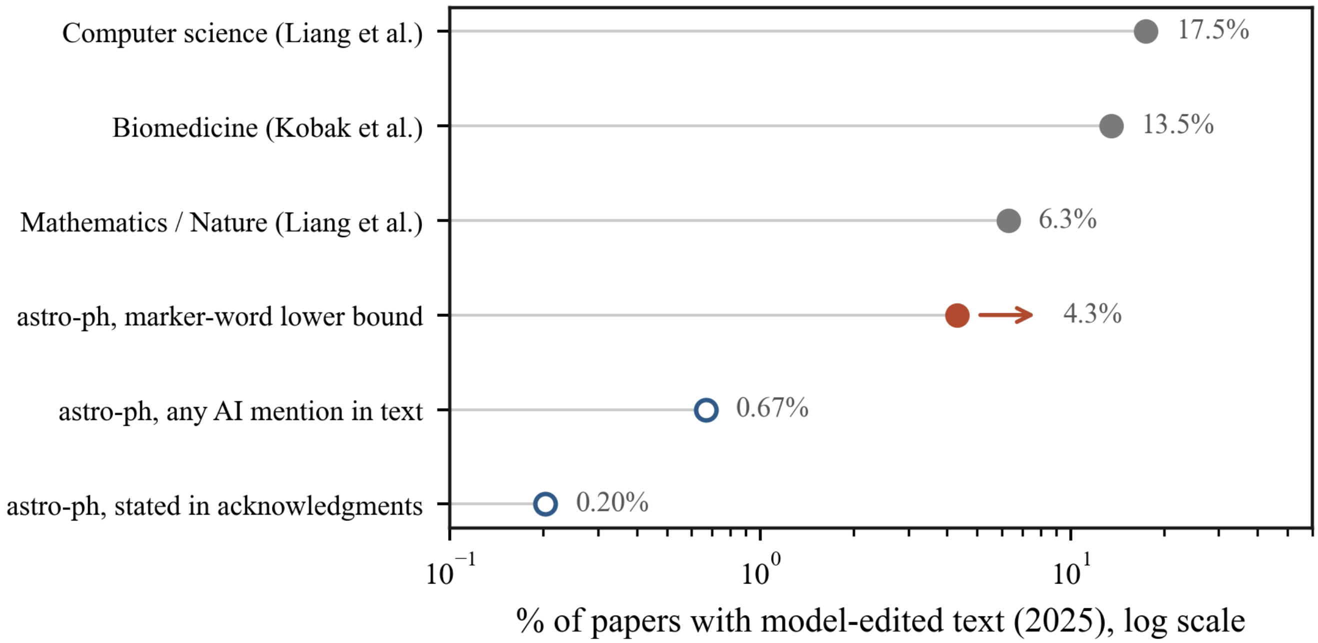

Figure 6. The model-touched fraction of 2025 papers across fields and detection routes, on a logarithmic axis. Grey points are published estimates for other fields (Liang et al. 2025; Kobak et al. 2025); the red point is our marker-word estimate for astro-ph, with the arrow marking it as a lower bound; open circles are the fractions of astronomy papers that mention or state model use. The gap between the red point and the open circles separates use from disclosure.

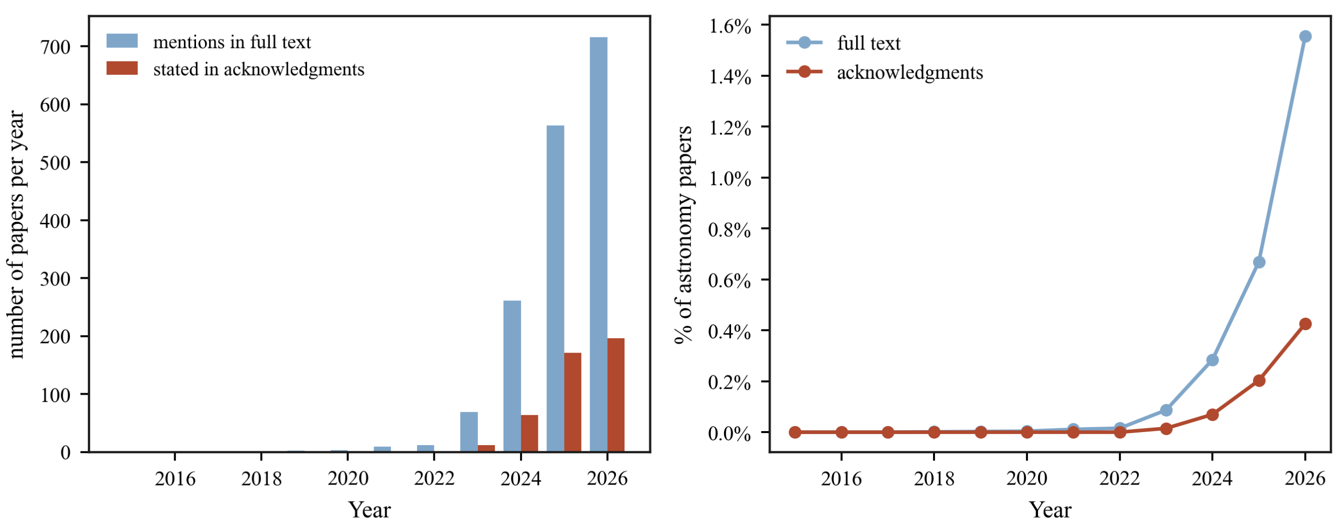

Figure 7. Left: number of astronomy papers per year that mention a model in full text (blue) or state model use in the acknowledgments (red); both counts are zero before 2023. Right: the same as a share of all astronomy papers, which stays below 1%.

3.6. *Source-text and citation audits*

The source-text search of Section 2.7 finds 3 matches in the 2,085 papers from 2024–2026, a rate of 0.14% (Wilson 95% interval 0.05–0.42%); two of the three are papers that study language models as a research topic rather than writing slips. The 1,244 pre-2023 papers give zero matches (95% upper limit 0.31%), so the false-positive rate is low. Figure 8 compares the four detection routes on the same corpus; they span two orders of magnitude. Direct source checks (0.14%) and stated disclosure (0.19%) find the least, and the word signal (4.3%, a lower bound) finds the most, so population statistics recover a factor of twenty more model use than direct evidence alone.

The citation audit of Section 2.8 finds that all 22,547 cited new-style arXiv identifiers (16,850 unique) resolve; none points to a paper that does not exist. Of the 2,500 sampled DOIs, Crossref returned 134 misses, and we inspected every one. Most are real records in registries that Crossref does not index: 87 arXiv DataCite identifiers, 22 Zenodo records, and 16 data-archive, regional-journal, and funder identifiers, of which four we confirmed directly through the global handle resolver. Four more are artifacts of our own extraction. The remaining five are wrong identifiers attached to real references: two DOIs in a single bibliography whose suffixes do not exist while every other field (authors, title, journal, volume, page, arXiv number) matches a real paper, two identifier schemes that are not registered DOIs (an ACM proceedings prefix and an ISBN written as a DOI), and one journal citation-template placeholder. We find no fabricated references, in the sense of a reference to a work that does not exist.

The null result carries statistical weight. Restricting to the 1,250 papers with at least one checkable arXiv identifier, zero flagged papers gives a 95% upper limit of 0.24% on the per-paper rate. At the early-2026 biomedical rate of one paper in 277 we would expect 4.5 flagged papers here, so that rate is excluded at 95% confidence ($p = 0.011$); the 2025 biomedical rate of one in 458 predicts 2.7 and is disfavored but not excluded ($p = 0.065$). Per cited identifier the limit is 1.3×10^{-4}. Two caveats bound the claim. The check tests whether a cited work exists, not whether it supports the statement it is attached to, and the wrong-suffix DOIs above show that identifier corruption occurs even when the reference is real. The test also covers papers that cite through arXiv identifiers, which skews toward recent, arXiv-native bibliographies (Figure 9).

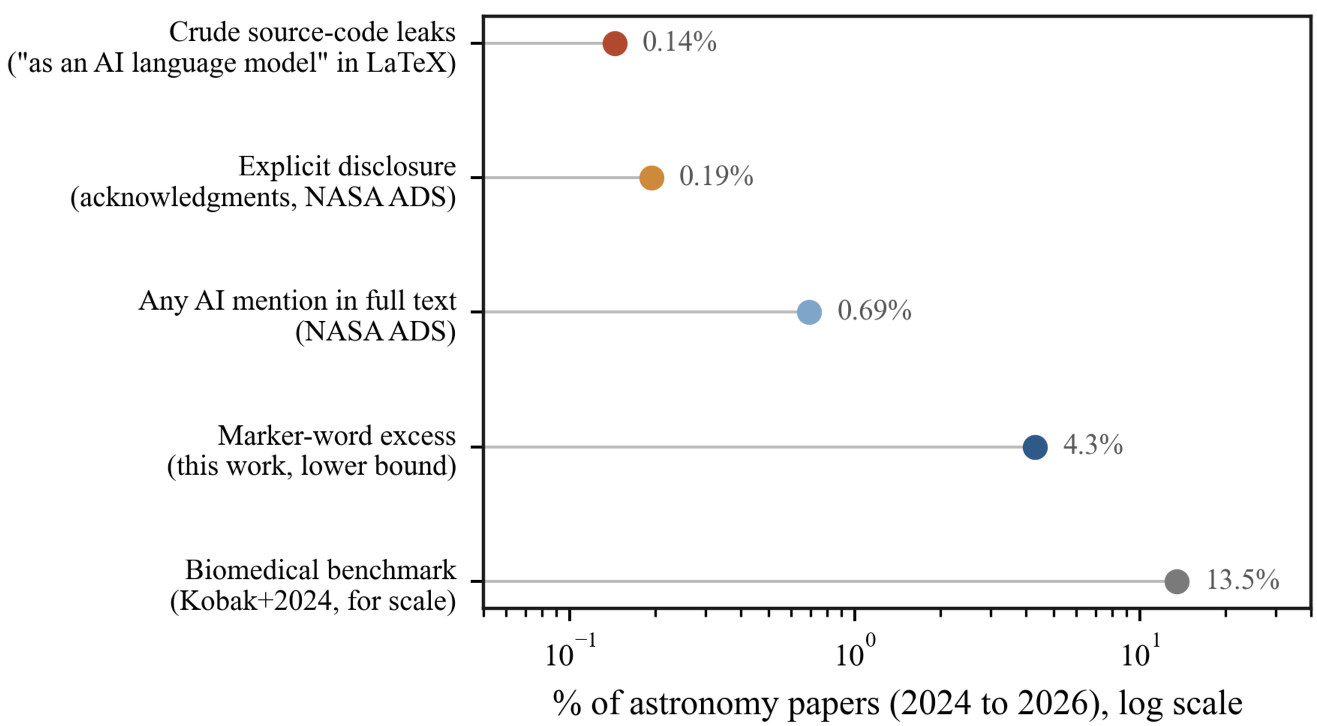

Figure 8. Four detection routes applied to the same 2024–2026 astronomy papers, on a logarithmic scale. Source-text matches and stated disclosure find the least; the word signal finds the most. The biomedical estimate (Kobak et al. 2025) is shown for scale.

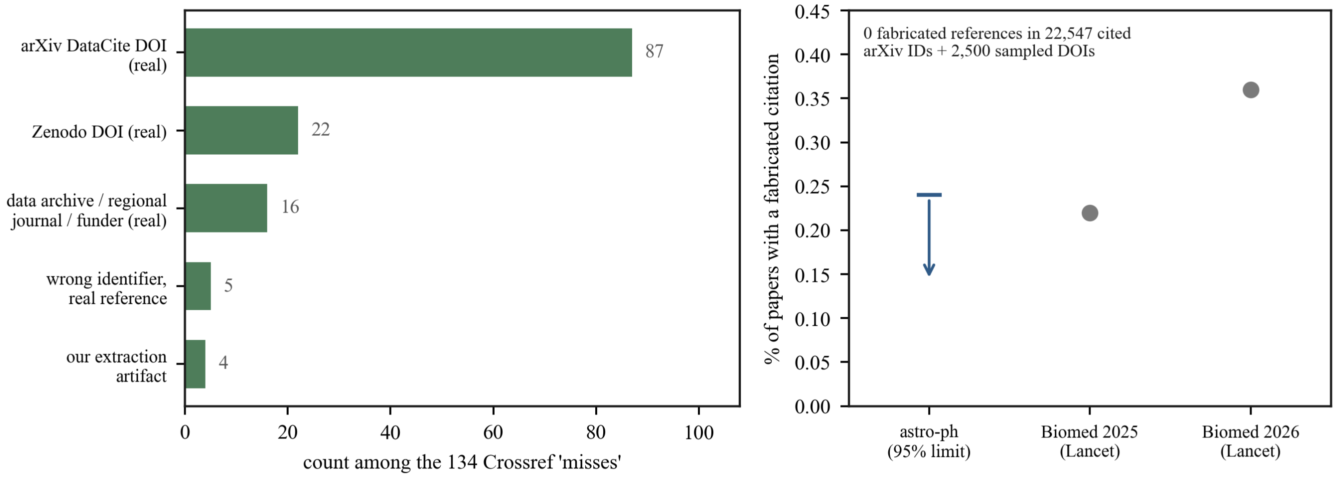

Figure 9. Left: classification, after manual inspection, of the 134 DOIs that a Crossref check could not resolve. None is a fabricated reference: most are real records in registries Crossref does not index, five are wrong identifiers attached to real references, and four are artifacts of our extraction. Right: the fabricated-citation rate; the astronomy value is the 95% upper limit from zero detections (arrow), against the biomedical audit of Topaz et al. (2026).

3.7. *Dependence on subfield and country*

The marker rate is not uniform across the field. By subfield (Figure 10), the Instrumentation and Methods category, which is closest to software and machine learning, has the highest rate throughout, reaching $7.8 \pm 1.3\%$ in 2025 (binomial 95%), while high-energy astrophysics has the lowest at $4.7 \pm 0.7\%$; the difference is large compared with its uncertainty. The ordering follows the pattern seen across science, where fields closer to computer science adopted these tools first.

By author country (Figure 11), obtained from the affiliation field, papers from countries where English is not a first language tend toward higher marker rates, with Iran (11.2%, Wilson 95% interval 8.6–14.6%, $n = 418$), China (7.6%, $n = 8{,}377$), and India (6.5%, $n = 3{,}179$) at the high end and Japan (4.2%) and Russia (1.3%) lower. This is consistent with the use of these tools as a writing aid by non-native English speakers, and it is a reason to treat a marker word as evidence of editing rather than of anything worse. Disclosure varies as well, and not along the same axis: papers with Polish (1.0%) and Japanese (0.8%) affiliations state model use most often, with the United States at 0.6%. Papers from China combine a high marker rate (7.6%) with one of the lowest disclosure rates (0.26%), the widest separation between the two quantities in the sample. The affiliation search counts every country on a paper, so large collaborations appear more than once, and the small-sample countries are noisy; these rates are indicative rather than precise.

3.8. *Promotional and hedging vocabulary*

The promotional word set rose by about 19% over the period and follows the marker rate closely ($r = 0.87$ over the twelve yearly points, $p = 2 \times 10^{-4}$); the hedging set did not move ($r = -0.21$; Figure 12). Promotional vocabulary therefore rose while stated caution did not. Both series trend with time, so part of the correlation comes from the shared calendar, and competition for attention raises promotional language on its own; we report this as a descriptive result rather than a causal claim.

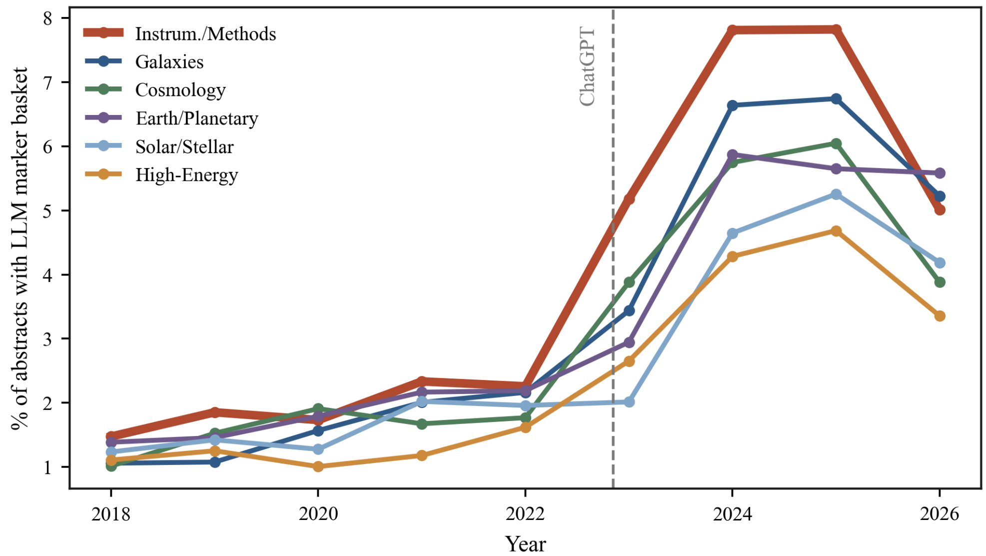

Figure 10. Marker rate by primary astro-ph subfield. Instrumentation and Methods has the highest rate throughout; high-energy astrophysics the lowest.

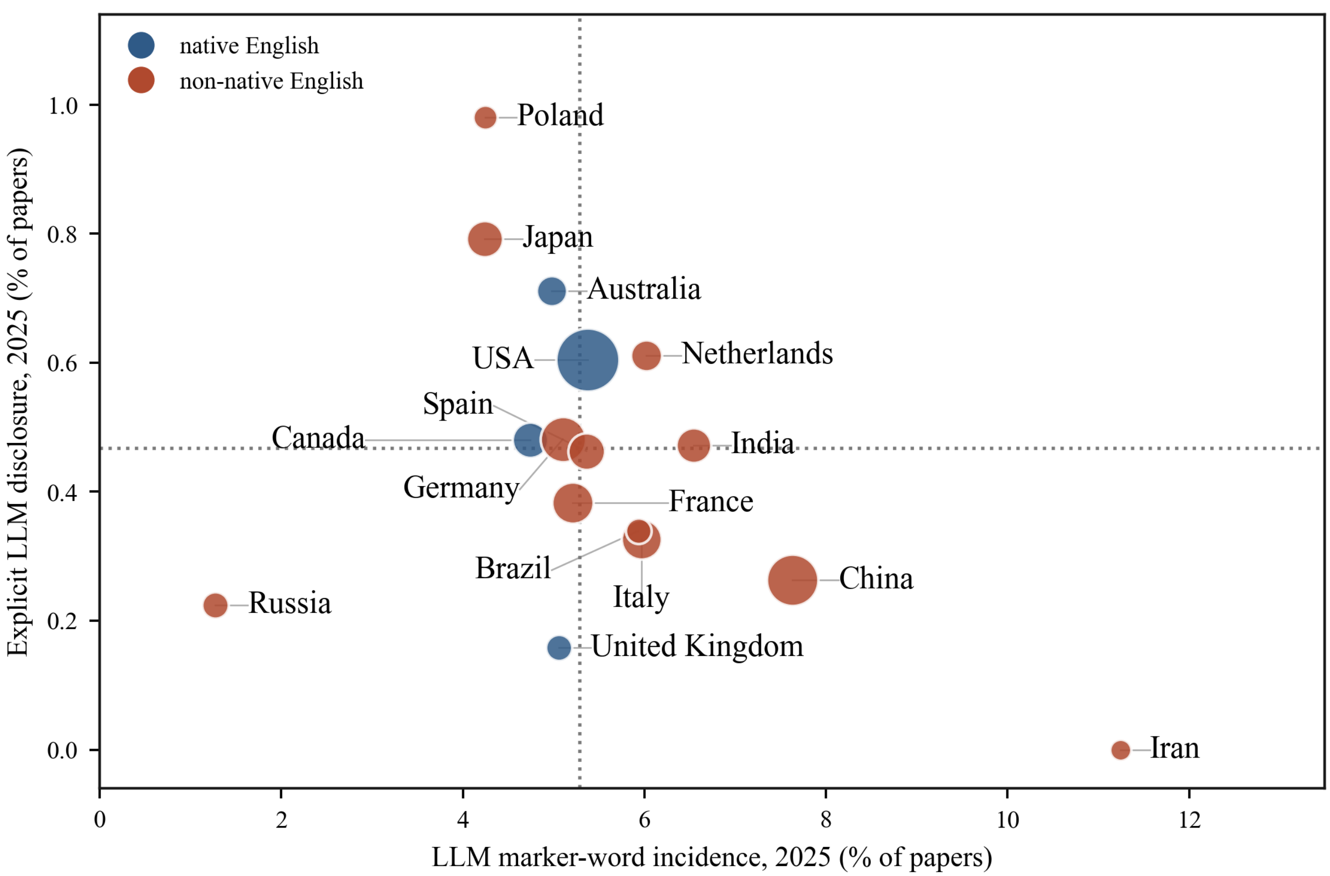

Figure 11. Marker rate against disclosure rate by author country in 2025. Blue points mark countries where English is a first language, and bubble size is the number of papers. Wilson 95% intervals on the marker rates are quoted in the text; the smallest sample shown is Iran with $n = 418$. The affiliation search counts every country on a paper, so large collaborations enter more than once.

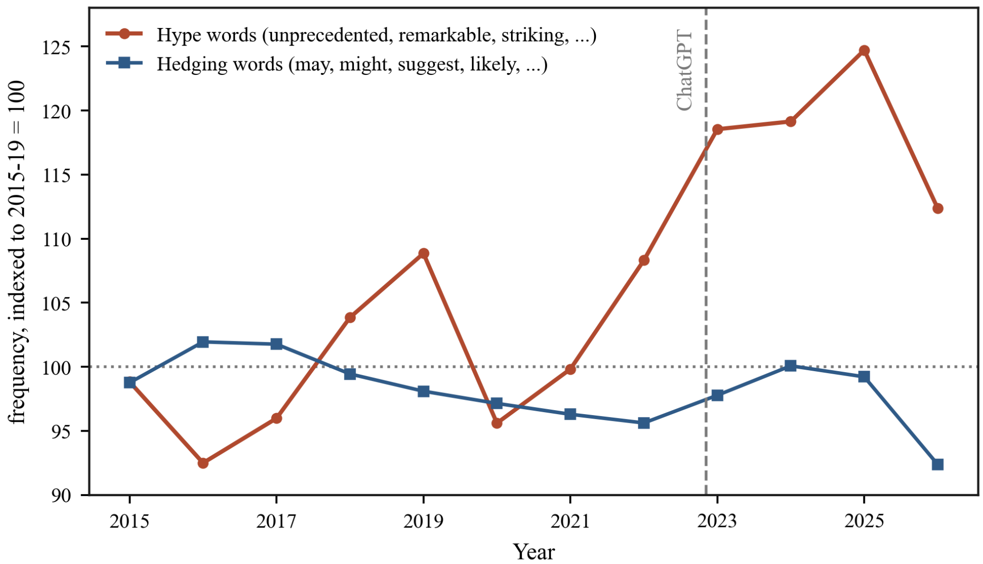

Figure 12. Promotional and hedging word frequency, indexed to the 2015–2019 mean. Promotional words rise together with the marker rate; hedging words stay flat.

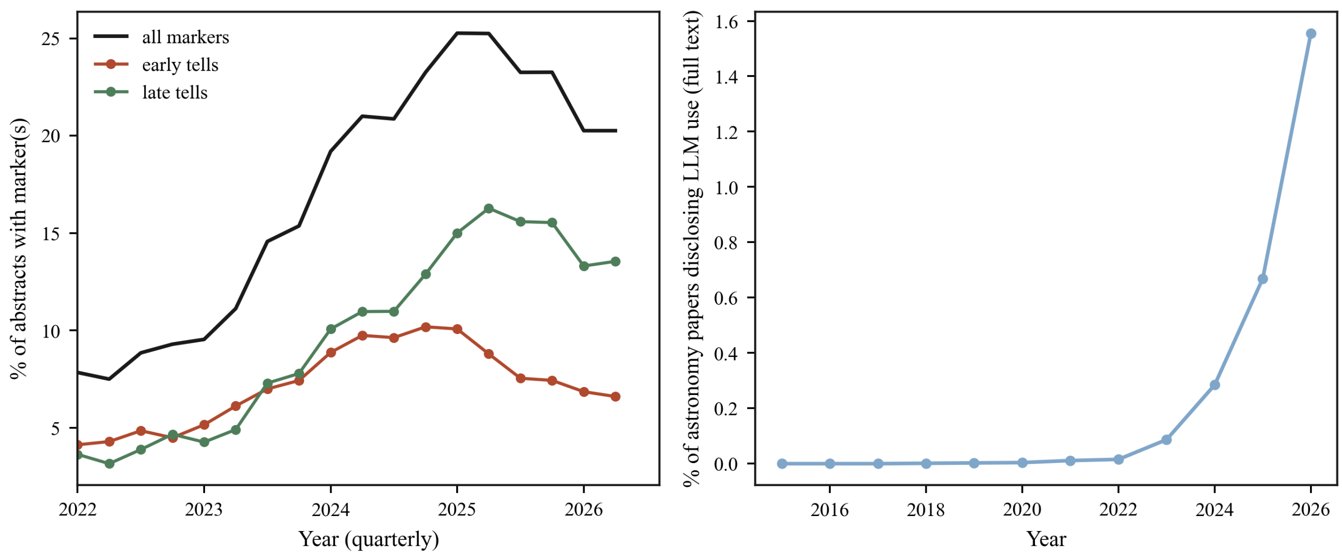

Figure 13. Left: the marker words weaken across the board in 2026, both the early markers that peaked first (`delve`, `intricate`, `pivotal`) and the later ones (`underscore`, `leveraging`). Right: over the same months, stated model use continues to rise.

3.9. *Time evolution of the word signal*

The marker rate declines slightly in 2026. Two observations argue against reading this as a decline in model use. Figure 13 places the marker rate next to the disclosure count over the same months: the marker words weaken across the board while stated model use continues to rise, passing the full 2025 count within half a year. The weakening therefore affects the word signal, not the underlying use. Newer models produce fewer of the known words, and authors edit the publicized ones out. Any word-based estimate is consequently a lower bound whose sensitivity declines with time, even as use grows.

4. DISCUSSION

Our main results are as follows.

1. The fraction of abstracts with a marker word rose from about 1.5% before 2022 to about 5.8% in 2025, while neutral control words stayed flat.

2. A conservative lower bound puts the model-touched fraction of 2025 abstracts near 4%; the true value is higher and is in the range measured for physics and mathematics.

3. Without an imported list, the data recover the same style words and phrases next to the genuinely new instrument and survey names, and the marker words occur together far more often than chance.

4. Stated disclosure stays below 1%, and direct source-text checks find near-zero pasted text; the word signal sees roughly two orders of magnitude more than either.

5. We find no fabricated citations, in contrast with a rising rate in biomedicine; the astronomy reference habit appears to protect the citation record.

6. The word signal weakens in 2026 even as disclosure rises, so word counts are a lower bound that will get harder to measure.

Taken together, these measurements give a consistent picture. Model-edited language is common and growing, stated use is rare, and the gap between the two is roughly a factor of twenty. The clustering of marker words within abstracts points to a single cause rather than unrelated drift, and direct checks of source text find almost nothing, so population statistics are the only reliable measurement. Astronomy so far avoids the fabricated-citation problem seen elsewhere, at a level we can bound below the early-2026 biomedical rate, most likely because of its shared reference database.

Several limits apply. First, a marker word measures model-edited language, not model-generated science; an abstract with `delve` in it is not fraudulent. Second, for many authors, and in particular for those writing in a second language, these tools lower a real barrier, so a marker word is better read as a sign of editing than of misconduct. Third, the word method has a short shelf life: as Section 3.9 shows, the known words fade as they become known, so a fixed list undercounts by more each year. Fourth, we read abstracts, not full papers, and abstracts are the most edited part of a paper. Finally, three uses that are easy to run together should be kept apart: a model as a copy editor, a model that drafts content, and a paper produced whole. The word counting mostly sees the first.

These results also bear on method. Word lists will keep losing power, so the more durable signals are the ones tied to content rather than style: the co-occurrence structure of Section 3.3, which is harder to remove by editing single words, and integrity checks such as the citation audit of Section 3.6, which astronomy can run more cleanly than most fields because of the NASA Astrophysics Data System.

The gap between use and disclosure has a bearing on refereeing. If a few percent of abstracts are model-edited while under one percent say so, then a referee or an editor cannot assume that a clean, fluent text was written without help, and a screening tool that flags marker words will misjudge honest authors far more often than it catches misuse, because the base rate of misuse is low and the base rate of editing is not. The same tool applied to referee reports would face the same problem. A disclosure norm, stated once at submission and kept simple, would remove the guesswork at low cost, and would separate the ordinary use of a model as an editor from the uses that raise real concern.

Why astronomy sits at the low end of the measured range is not settled by our data, but three contributions are plausible. The subfield ordering of Section 3.7 shows that adoption tracks proximity to computer science, and most of astronomy is farther from that boundary than the fields at the top of the range. Large collaboration papers, common in astronomy, pass through many hands and style conventions that dilute any single writer's tooling. And part of the gap is definitional: our floor-type estimator would report a lower number than a mixture fit even on identical text, as noted in Section 2.5.

Several extensions follow directly. The same measurements can be run on full text rather than abstracts, where the signal should be weaker per word but far larger in volume, and on other reviewed text in the field, such as observing proposals, where the archives permit. The vocabulary should be re-derived at each new harvest rather than frozen, since Section 3.9 shows a fixed list losing sensitivity within about two years.

There is also a feedback loop to consider. Models trained on their own output degrade (Shumailov et al. 2024), and model-mediated text narrows the diversity of claims it carries (Wright et al. 2025). As a growing share of the published corpus is model-edited, the next generation of models, and any astronomy-specific models trained on astro-ph text (Perkowski et al. 2024), will learn from text that already carries the style. Word-frequency screening will then measure the training data as much as the writing, and its output will drift. The signals that do not depend on a fixed word list, such as the co-occurrence structure and the citation checks, are the ones that will retain their meaning.

Whether astronomy should agree on a disclosure norm now, before a single case forces one, is a question the field can still answer on its own terms.

ACKNOWLEDGMENTS

We thank Yuan-Sen Ting for suggesting this work be written up and for pointing to the co-occurrence measurement. This research used the arXiv API and the NASA Astrophysics Data System. All data and code are available at `github.com/seratsaad/llm-in-astro-ph`.

Software: NumPy, pandas, Matplotlib, PyArrow

REFERENCES

Astarita, S., Kruk, S., Reerink, J., & Gómez, P. 2024, arXiv e-prints, arXiv:2406.17324

Ciucă, I., & Ting, Y.-S. 2023, Research Notes of the AAS, 7, 193, doi: 10.3847/2515-5172/acf85f

Else, H. 2023, Nature, 613, 423, doi: 10.1038/d41586-023-00056-7

Fouesneau, M., Momcheva, I. G., Chadayammuri, U., et al. 2024, arXiv e-prints, arXiv:2409.20252

Geng, M., & Trotta, R. 2025, Findings of the Association for Computational Linguistics: ACL 2025, 12689, doi: 10.18653/v1/2025.findings-acl.657

Glynn, A. 2024, arXiv e-prints, arXiv:2411.15218

Gray, A. 2024, arXiv e-prints, arXiv:2403.16887

—. 2025, arXiv e-prints, arXiv:2512.01560

Iyer, K. G., Yunus, M., O'Neill, C., et al. 2024, ApJS, 275, 38, doi: 10.3847/1538-4365/ad7c43

Juzek, T. S., & Ward, Z. B. 2024, Proceedings of COLING, arXiv:2412.11385

Kobak, D., González-Márquez, R., Horvát, E.-Á., & Lause, J. 2025, Science Advances, 11, doi: 10.1126/sciadv.adt3813

Kurtz, M. J., Eichhorn, G., Accomazzi, A., et al. 2000, A&AS, 143, 41, doi: 10.1051/aas:2000170

Liang, W., Yuksekgonul, M., Mao, Y., Wu, E., & Zou, J. 2023, Patterns, 4, 100779, doi: 10.1016/j.patter.2023.100779

Liang, W., Izzo, Z., Zhang, Y., et al. 2024, Proceedings of ICML, arXiv:2403.07183

Liang, W., Zhang, Y., Wu, Z., et al. 2025, Nature Human Behaviour, 9, 2599, doi: 10.1038/s41562-025-02273-8

Nature. 2023, Nature, 613, 612, doi: 10.1038/d41586-023-00191-1

Perkowski, E., Pan, R., Nguyen, T. D., et al. 2024, Research Notes of the AAS, 8, 7, doi: 10.3847/2515-5172/ad1abe

Richardson, R. A. K., Hong, S. S., Byrne, J. A., Stoeger, T., & Amaral, L. A. N. 2025, PNAS, 122, doi: 10.1073/pnas.2420092122

Shumailov, I., Shumaylov, Z., Zhao, Y., et al. 2024, Nature, 631, 755, doi: 10.1038/s41586-024-07566-y

Stokel-Walker, C., & Van Noorden, R. 2023, Nature, 614, 214, doi: 10.1038/d41586-023-00340-6

Thelwall, M., & Kousha, K. 2026, arXiv e-prints, arXiv:2604.07565

Ting, Y.-S., Nguyen, T. D., Ghosal, T., et al. 2024, arXiv e-prints, arXiv:2407.11194

Topaz, M., Roguin, N., Gupta, P., Zhang, Z., & Peltonen, L.-M. 2026, The Lancet, 407, 1779, doi: 10.1016/s0140-6736(26)00603-3

Van Noorden, R., & Perkel, J. M. 2023, Nature, 621, 672, doi: 10.1038/d41586-023-02980-0

Wright, D., Masud, S., Moore, J., et al. 2025, arXiv e-prints, arXiv:2510.04226

APPENDIX

A. THE MARKER BASKET

Table 3 lists the full 38-word basket used for the lower bound, with document frequencies in the baseline and recent windows. The control set is `observed`, `measured`, `obtained`, `presented`, `galaxy`, `stellar`, `sample`, `temperature`, `redshift`, and `spectra`.

Table 3. The marker basket: document frequency (per cent of abstracts) in 2018–2021 and 2024–2026.

word	2018–21	2024–26	ratio
leveraging	0.131	1.224	9
underscore	0.025	0.585	23
pivotal	0.082	0.468	6
underscores	0.030	0.449	15
intricate	0.072	0.389	5
encompassing	0.133	0.352	3
elucidate	0.181	0.340	2
showcase	0.124	0.305	2
underscoring	0.010	0.278	27
plethora	0.137	0.132	1
unraveling	0.046	0.120	3
showcasing	0.012	0.118	10
unravel	0.094	0.115	1
delve	0.016	0.113	7
showcases	0.025	0.107	4
elucidating	0.042	0.094	2
nuanced	0.018	0.081	5
seamlessly	0.034	0.073	2
noteworthy	0.061	0.071	1
delves	0.001	0.068	45
seamless	0.024	0.068	3
realm	0.073	0.060	1
multifaceted	0.013	0.047	3
myriad	0.051	0.043	1
garnered	0.012	0.041	3
unravelling	0.015	0.041	3
meticulous	0.009	0.032	4
meticulously	0.009	0.032	4
intricacies	0.013	0.028	2
showcased	0.003	0.026	9
tapestry	0.001	0.017	11
boasts	0.007	0.015	2
delving	0.003	0.015	5
realms	0.010	0.006	1
commendable	0	0.004	···
garner	0	0.004	···
testament	0.001	0.004	3
garners	0	0.002	···